Unit I: Finite Automata

Finite Automata: (05 hrs)

- Automaton as a model of computation,
- Symbols,
- Alphabets,
- Strings,
- Languages,
- Deterministic Finite Automata (DFA),
- Nondeterministic finite Automata (NFA),
- State Minimization algorithm,
- NFA with epsilon transition,
- pumping lemma

Basic Terminologies of Theory of Computation:

- Symbol
- Alphabet
- String
- Language

Symbols

Symbols are called as a characters. it is the smallest building block, which can be any alphabet, letter, or picture.

or

Symbol is an atomic unit, such as a digit, character, lowercase letter, etc.

For example,

a,b,c,……z
0,1,2,……..9
+,-,*,%,……special characters($,#,@).

Alphabet

The set of characters is called as an alphabet .

An alphabet is a <u>finite</u>, <u>non-empty</u> set of symbols.

It is denoted by Σ.

For Example,

Σ ={0,1} set of binary alphabets.

Σ ={a,b,c,……..,z} set of all lower case letters.

Σ ={A,B,C,……Z} set of all upper case letters.

Σ ={+,&,%,…….} set of all special characters.

String or Word

A string is a finite set sequence of symbols chosen from some alphabets.

<u>Empty string</u> is the string with zero occurrence of symbols, represented as ε.

A string is generally denoted as w and the length of a string is denoted as |w|.

For example,

If Σ={0,1} is set of the binary alphabet then string

w=00011001 and |w|=8

String

For example,

If $\Sigma=\{a,b,c,d\}$ is set of alphabet then string

$w=aabbcabcd$ and $|w|=9$

If, $w = 0110$

$y = baa$

$x = aabcaa$

$z = 111$. Then,

Concatenation – $wz = 0110111$

Special string –$s= \{\epsilon\}$

Length – $|w| = 4$, $|s| = 0$, $|x| = 6$, $|wz|=7$

Reversal – $y^R = aab$

Find Number of Strings (of length 2) that can be generated over the alphabet $\Sigma=\{a, b\}$:

a	a
a	b
b	a
b	b

Length of String $|w| = 2$

Number of Strings = 4

For alphabet $\Sigma=\{a, b\}$ with length n, number of strings can be generated = 2^n.

Length of string

It is the number of symbols in the string or word. It is denoted by |w|.

For example,

w=01011001 from binary alphabet Σ={0,1}

|w| = 8

X= abbaddabba from binary alphabet Σ={a,b}

|X| = 10

Language

A language is a set of strings from some alphabet (finite or infinite).

For example,

If alphabet Σ=`{a,b}then`

1) **Language over these alphabets is L={a,b,aa,ab,ba,bb,aaa,bbb,abb,aab,baa,bba,aba,bab,aaaa,bbbb,aabb,bbaa……..} (infinite)**
2) **Language over these alphabets is having length 3 is L={aaa,bbb,abb,aab,baa,bba,aba,bab} Finite**

Language

Some special sets of strings are as follows –

{} The empty set/language, containing no string, also called as ϕ Phi.

{ϵ} A language containing one string, the empty string.

Σ^* All strings of symbols from Σ including ϵ

$\Sigma^+ = \Sigma^* - \{\epsilon\}$ or $\Sigma^* = \Sigma^+ + \{\epsilon\}$

For example,

$\Sigma = \{0, 1\}$

$\Sigma^* = \{\epsilon, 0, 1, 00, 01, 10, 11, 000, 001, \ldots\}$

$\Sigma+ = \{0, 1, 00, 01, 10, 11, 000, 001, .\}$

Language

$\Sigma = \{0, 1\}$

$L = \{x \mid x \text{ is in } \Sigma^* \text{ and } x \text{ contains an even number of } 0's\}$

$\Sigma = \{0, 1, 2, \ldots\ldots 9\}$

$L = \{x \mid x \text{ is in } \Sigma^* \text{ and } x \text{ forms a infinite length decimal numbers}\}$

$= \{0, 10, 9,3,2,6, \ldots\ldots\ldots\ldots\}$

Set,Subsets,Superset

A **set** is a collection of objects or elements, grouped in the curly braces.

Ex. A= {a,b,c,d} , B={ 1,2,3,4,5}

Subsets are a part of one of the mathematical concepts called Sets.

Ex.

$A=\{2,4,6,8,10,12,14,16,18,20\}$

$B=\{2,4,6\}$

$B \subseteq A$

If a set A is a collection of even number and set B consists of {2,4,6}, then B is said to be a subset of A, denoted by $B \subseteq A$ and A is the **superset** of B.

All Subsets of a Set

The subsets of any set consists of all possible sets including its elements and the null set. Let us understand with the help of an example.

Example: Find all the subsets of set A = {1,2,3,4}

Solution: Given, A = {1,2,3,4}

Subsets = {}

{1}, {2}, {3}, {4},

{1,2}, {1,3}, {1,4}, {2,3},{2,4}, {3,4},

{1,2,3}, {2,3,4}, {1,3,4}, {1,2,4}

{1,2,3,4}

Types of Subsets

Subsets are classified as

- Proper Subset
- Improper Subsets

A proper subset is one that contains a few elements of the original set whereas an improper subset, contains every element of the original set along with the null set.

For example, if set A = {2, 4, 6}, then,

Number of subsets: {2}, {4}, {6}, {2,4}, {4,6}, {2,6}, {2,4,6} and Φ or {}.

Proper Subsets: {}, {2}, {4}, {6}, {2,4}, {4,6}, {2,6}

Improper Subset: {2,4,6}

Proper Subsets

Set A is considered to be a proper subset of Set B if Set B contains at least one element that is not present in Set A.

Example: If set A has elements as {12, 24} and set B has elements as {12, 24, 36}, then set A is the proper subset of B because 36 is not present in the set A.

Proper Subset Symbol

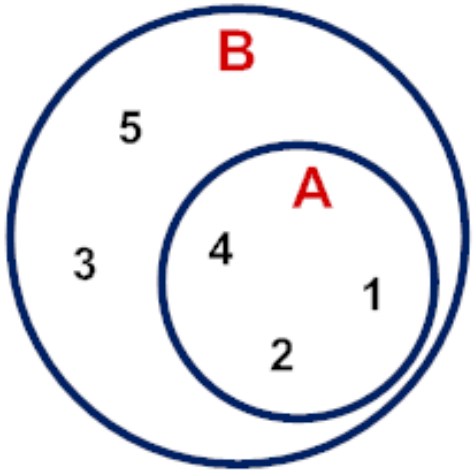
A proper subset is denoted by ⊂ and is read as ‘is a proper subset of’. Using this symbol, we can express a proper subset for set A and set B as;

$$A \subset B$$

Example 1: Given $A = \{1, 2, 4\}$ and $B = \{1, 2, 3, 4, 5\}$, what is the relationship between these sets?

We say that ***A* is a subset of *B***, since every element of A is also in B. This is denoted by:

A <u>Venn diagram</u> for the relationship between these sets is shown to the right. $A \subset B$



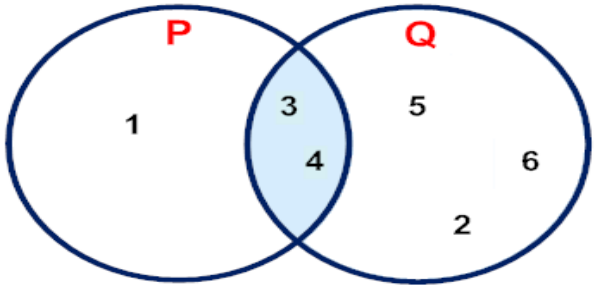
Not subset of

Given $P = \{1, 3, 4\}$ and $Q = \{2, 3, 4, 5, 6\}$, what is the relationship between these sets?

We say that P is not a subset of $\boldsymbol{Q}$ since not every element of P is not contained in Q. For example, we can see that $1 \notin Q$. The statement *"P is not a subset of Q"* is denoted by:

Note that these sets do have some elements in common. The intersection of these sets is shown in the Venn diagram below

$$P \not\subset Q$$



Proper Subset

Subset Example : Conclude whether X is a subset of Y.

X = {All writing material in a stationary workshop}, Y = {Pencils}

Set X involves pens, sketch pens, markers, pencils, notepads, etc. Whereas set Y only carries pencils. So we cannot state that all X's elements are present in Y, which is a requirement for X to be a subset of Y. In this particular case, we can state that Y is a subset of X, but X is not a subset of Y.

Subset Example : Determine whether A is a subset of B.

A = {Toyota}, B = {All brands of cars}

Set B covers all brands of cars; Maruti Suzuki, Hyundai, Toyota Mahindra, Tata Motors, Mercedes Benz, etc. Moreover, A is a set of Toyota. Then we can say that all elements of A are incorporated into B. Hence, A is a subset of B.

How many subsets and proper subsets does a set have?

If a set has “n” elements, then the number of subset of the given set is 2^n and the number of proper subsets of the given subset is given by 2^n-1.

Consider an example, If set A has the elements, A = {a, b}, then the proper subset of the given subset are { }, {a}, and {b}.

Here, the number of elements in the set is 2.

We know that the formula to calculate the number of proper subsets is $2^n – 1$.

$= 2^2 – 1$

$= 4 – 1$

$= 3$

Thus, the number of proper subset for the given set is 3 ({ }, {a}, {b}).

Improper Subset

A subset which contains all the elements of the original set is called an improper subset. It is denoted by $\subseteq$.

For example: Set P ={2,4,6} Then, the subsets of P are;

{}, {2}, {4}, {6}, {2,4}, {4,6}, {2,6} and {2,4,6}.

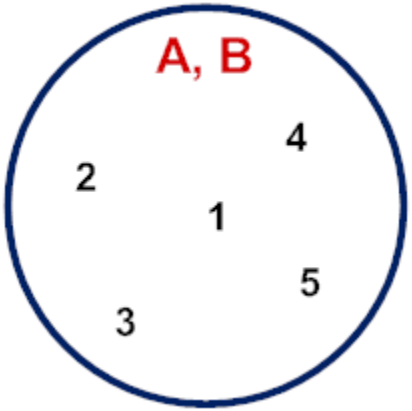
Where, {}, {2}, {4}, {6}, {2,4}, {4,6}, {2,6} are the proper subsets and {2,4,6} is the improper subsets. Therefore, we can write $\{2,4,6\} \subseteq P$.

Note: The **empty set** is an **improper subset of itself** (since it is equal to itself) but it is a *proper subset of any other set*.

Equal sets

Given $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 1, 2, 5, 4\}$, what is the relationship between A and B?

Analysis: Recall that the order in which the elements appear in a set is not important. Looking at the elements of these sets, it is clear that: $A=B$ $B \subset A$ $A \subset B$



Power Set

The **power set** is said to be the collection of all the subsets. It is represented by P(A).

If A is set having elements {a, b}. Then the power set of A will be;

P(A) = {∅, {a}, {b}, {a, b}}

Properties of Subsets

Some of the important properties of subsets are:

- Every set is a subset of the given set itself. It means that $X \subset X$ or $Y \subset Y$, etc
- We can say, an empty set is considered as a subset of every set.
- X is a subset of Y. It means that X is contained in Y
- If a set X is a subset of set Y, we can say that Y is a superset of X

Automaton as a model of computation

Automaton: Automatons are abstract models of machines that perform computations on an input by moving through a series of states or configurations.

Or

An automaton (automata in plural) is an abstract self-propelled computing device which follows a predetermined sequence of operations automatically.

At each state of the computation, a transition function determines the next configuration on the basis of a finite portion of the present configuration.

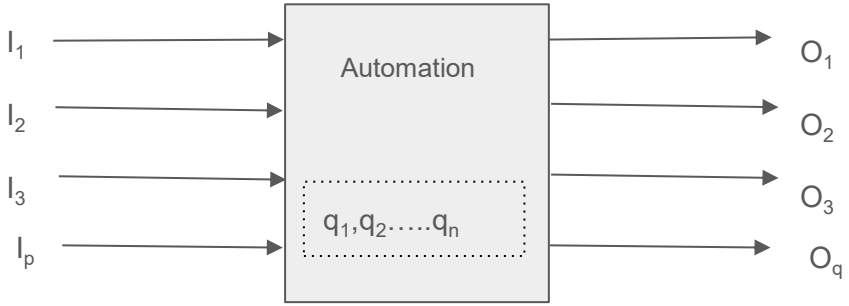
Finite automation:

An automaton with a finite number of states is called a Finite Automaton (FA) or Finite-State Machine (FSM).

The automaton consists of states (represented by circles) and transitions (represented by arrows). As the automaton sees a symbol of input, it makes a transition (or jump) to another state, according to its transition function, which takes the previous state and current input symbol as its arguments.

Definition of Automation

An Automation is defined as a system where energy, materials and information are transformed or transmitted and used for performing some functions without direct participation of man .



Model of discrete Automation

The Characteristic of automation

1. Input:
2. Output :
3. States :
4. State relation :
5. Output Relation:

Finite State Machine Formal Definition

A Finite automation can be represented by 5- tuples (Q,Σ,δ,q_0,F) , where

Q is a finite set of states

Σ is a finite set of An input alphabet

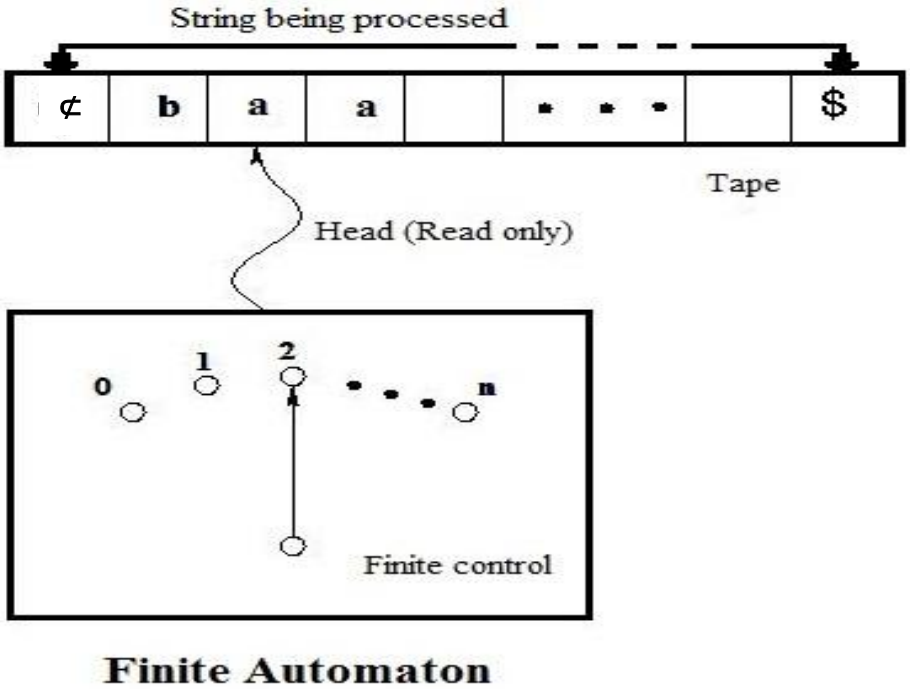
δ is a transition function

Q_0 is a start state

$F \subseteq Q$ is a set of final states

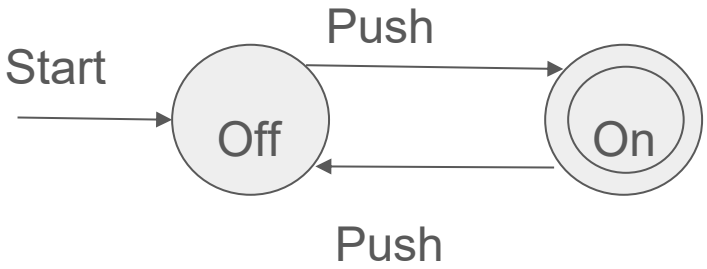
“Final” and “accepting” are synonyms.

Working of FSM



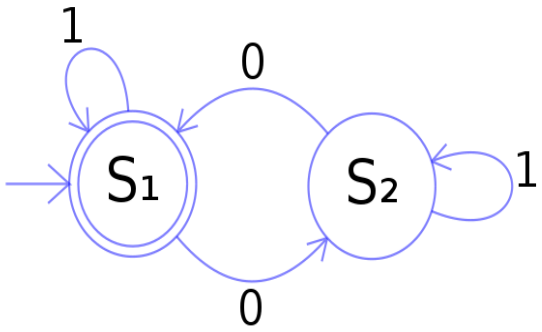
1. Input tape
2. Read head
3. Finite control

State Transition Diagram



Finite automation modeling an On/Off Switch

Finite State Machine(Transition System)



- The simplest representation by a **graph**.
  - Nodes = states.
  - Directed arcs indicate state transitions.
  - Labels on arcs tell what causes the transition.

The automaton described in this state diagram starts in state S_1, and changes states following the arrows marked 0 or 1 according to the input symbols as they arrive. The double circle marks S_1 as an accepting state. Since all paths from S_1 to itself contain an even number of arrows marked 0, this automaton accepts strings containing even numbers of 0s.

Finite Automata (FA)

The applications of Finite Automata are as follows −

- Design of the lexical analysis of a compiler.
- Recognize the pattern by using regular expressions.
- Use of the Mealy and Moore Machines for designing the combination and sequential circuits.
- Helpful in text editors.
- Used for spell checkers.